Do Now · retrieval

Programming Techniques · Value and Reference

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a local variable?

Q2

Why use a method?

Q3

What does an accumulator store?

Q4

What is a hash collision?

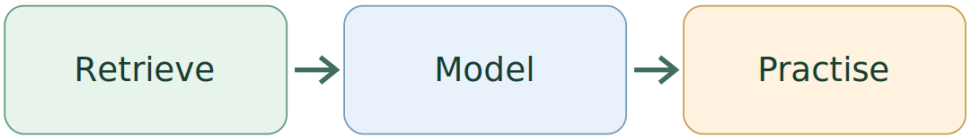
New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Explain the important decision — An int parameter normally receives a copied number. A Box parameter normally receives a copied reference to the same object. ref instead aliases the caller variable, allowing that variable to be reassigned.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

by value — Copies a value to a parameter

ref — Allows changes to caller variable

reference — Identifies an object

argument — Value provided to a call

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Compare changes to a copy and an object

Follow the worked method

1. Copy an integer value — Start x=5; CopyInt(x) changes only local n to 9. The caller x remains 5. This is pass-by-value.

2. Copy a reference; mutate one object — Start item→Box A(Value=5), alias=item. Mutate(item) copies the reference, then changes Box A.Value to 9. Both item and alias still refer to Box A.

3. Reassign the caller through ref — Replace(ref item) makes caller item refer to new Box B(Value=20). alias still refers to Box A(Value=9). Without ref, assigning a new Box to local p would leave item unchanged.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Discuss, then respond
If a method takes ref int n and adds 2, starting 3, what is the caller value?

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Apply independently

Explain why changing a normal int parameter does not change the original. [2]

Question 2

Apply independently

Write a C# method header to increment an integer by reference. [2]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Apply independently

Distinguish passing an object reference by value from using ref. [3]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

True or false: a reference-type argument passed normally allows the method to replace the caller's variable.

Use key vocabulary from memory

1. Explain by value in your own words.

2. Explain ref in your own words.

3. Explain reference in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.